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IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID: 25-26J-264

1. Topic (12 words max)

A smart mushroom incubation system for enhanced yield and sustainability in small-scale farming.

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Mushroom cultivation, particularly of oyster mushrooms (Pleurotus ostreatus), has become an increasingly important livelihood for small-scale farmers in Sri Lanka due to its nutritional value, high market demand, and relatively simple growing requirements (Bandara et al., 2020). However, despite its potential, traditional cultivation methods still dominate the sector. These methods rely heavily on manual labor and subjective assessments such as judging temperature by physical sensation or estimating humidity levels by touch which often lead to inconsistent yields and increased vulnerability to crop loss (Jayasinghe et al., 2018).

One of the core issues is the lack of precise environmental monitoring and control. Factors such as temperature, humidity, CO_2 concentration, and light intensity play a critical role in mushroom growth and quality (Chang & Miles, 2004). Inaccurate or delayed responses to fluctuations can result in suboptimal growth conditions, reduced yields, and increased susceptibility to pests and diseases. For example, CO_2 buildup or improper humidity can stunt mushroom development or encourage contamination, but these issues are often detected only after significant damage has occurred (Pathmashini et al., 2008).

Additionally, pest infestations and contamination are major threats that are typically identified too late, leading to substantial crop losses and financial setbacks for farmers (Bandara et al., 2020). The labor-intensive nature of manual monitoring also limits scalability and sustainability, as farmers must devote significant time and energy to routine checks, often with limited technical knowledge.

Given these challenges, there is a clear need for innovative solutions that can enhance both yield and sustainability in small-scale mushroom farming. Our research addresses this gap by proposing a smart mushroom incubation system equipped with IoT-enabled sensors and automated controls. This system aims to provide real-time monitoring and optimization of key environmental parameters, early detection of anomalies, and data-driven insights for continuous improvement. By reducing reliance on subjective assessments and manual interventions, this approach has the potential to transform mushroom cultivation into a more predictable, efficient, and sustainable enterprise for small-scale farmers in Sri Lanka.

References

[1] A. R. Bandara, S. Rapior, J. D. Bhat, and P. Kakumyan, "Current status of mushroom production in Sri Lanka and future prospects," *Mycosphere*, vol. 11, no. 1, pp. 1–14, 2020. [Online]. Available: https://doi.org/10.5943/mycosphere/11/1/1

[2] C. Jayasinghe, D. S. Wijesekara, and A. Gunasekara, "Challenges and prospects of mushroom cultivation in Sri Lanka," *Tropical Agricultural Research*, vol. 29, no. 3, pp. 345–353, 2018. [Online]. Available: https://doi.org/10.4038/tar.v29i3.8324

[3] S. T. Chang and P. G. Miles, *Mushrooms: Cultivation, Nutritional Value, Medicinal Effect, and Environmental Impact*, 2nd ed. CRC Press, 2004.

[4] L. Pathmashini, V. Arulnandhy, and W. Wilson, "Cultivation of oyster mushroom (Pleurotus ostreatus) on sawdust," *Ceylon Journal of Science (Biological Sciences)*, vol. 37, no. 2, pp. 177–182, 2008. [Online]. Available: https://doi.org/10.4038/cjsbs.v37i2.5033

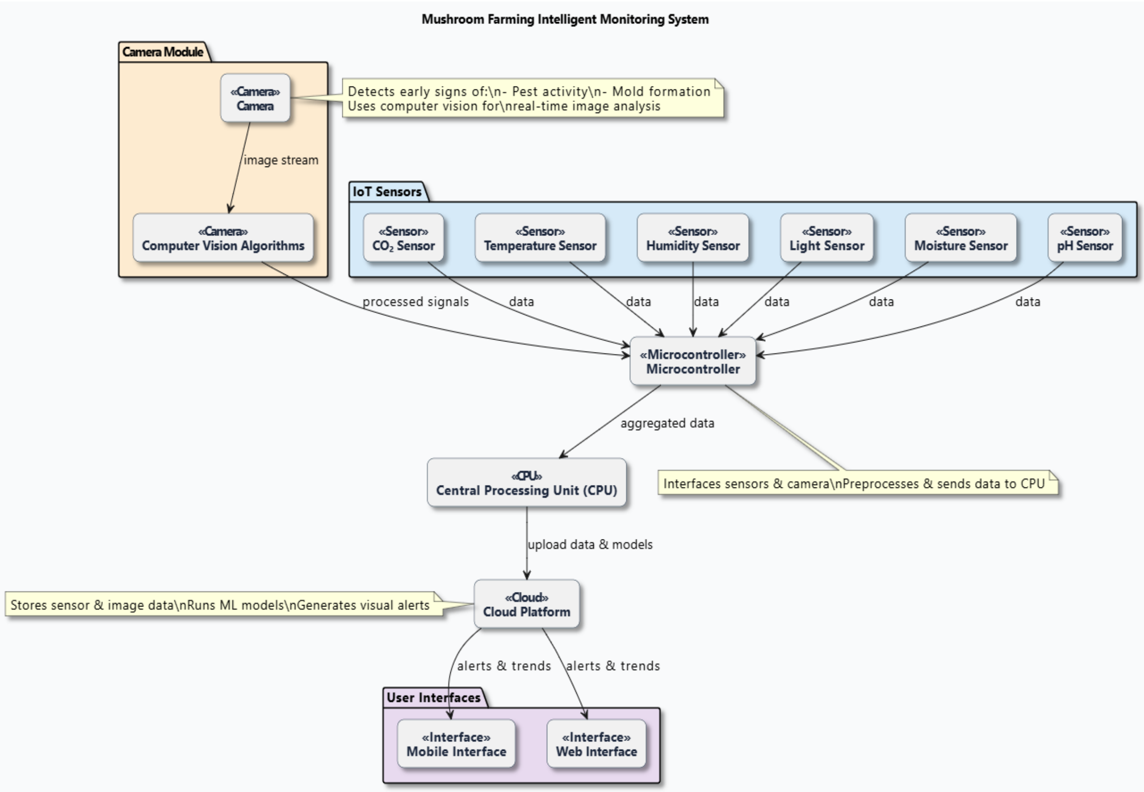
6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

Our proposed solution presents an intelligent monitoring system tailored for mushroom farming in Sri Lanka, addressing common issues such as pest infestations, contamination, poor environmental control, and inconsistent moisture management. The system integrates IoT-based sensors with machine learning techniques to monitor CO_2 levels, temperature, humidity, light, moisture, and pH in real time. A camera-based module, enhanced by computer vision algorithms, identifies early signs of pest activity and mold formation on mushroom bags.

The conceptual framework connects all hardware components sensors, cameras, and microcontrollers to a central processing unit, which transmits data to a cloud platform. A mobile/web interface displays visual alerts and environmental trends, supporting timely interventions.

While the current system focuses on a specific mushroom species, future research could extend to other species with similar growth characteristics, such as abalone mushrooms, which are also vulnerable to green mold. The limited dataset size, particularly for image-based analysis, poses scalability challenges. Factors like lighting variation, camera angle, and shadow interference may reduce the accuracy of visual monitoring.

Additionally, differences in farm layouts, equipment, and hygiene practices during data collection may impact model generalizability across farms. Addressing these variations through larger, more diverse datasets and adaptable models will be essential in future developments. Despite current limitations, this solution offers a promising foundation for precision farming and sustainable mushroom cultivation.



Conceptual Architecture with IoT, ML, and Cloud Integration

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

This research project calls for a multidisciplinary approach that brings together agricultural knowledge, embedded systems, and IoT automation—specifically focused on oyster mushroom cultivation in Sri Lanka. Our team needs a solid understanding of the biological and environmental needs of Pleurotus ostreatus, such as maintaining optimal temperature (25–30°C), relative humidity (70–90%), proper CO_2 levels, and suitable substrate conditions (including pH, moisture, and nutrient balance).

One of our key responsibilities is to collect primary data directly from small-scale oyster mushroom farmers across Sri Lanka. This involves visiting farms, conducting structured interviews, and observing current cultivation practices. We'll be gathering insights on common challenges like pest infestations, contamination issues, and inconsistent watering techniques, as well as understanding how farmers currently manage their growing environments. This information will directly inform the design and configuration of our automated system.

On the technical side, the project requires expertise in integrating environmental sensors (for temperature, humidity, CO_2, light, moisture, and pH), programming microcontrollers such as the ESP32 or Arduino, and developing simple, low-cost hardware solutions that can be scaled for rural use. Understanding the basic biology of microbial threats and how to detect contamination early is also essential, given how quickly infections can impact mushroom yields.

The data we collect will not only help shape the system's functions but also serve as a baseline for validating its performance—allowing us to compare traditional growing methods with our smart, automated alternative. We also need to be mindful of the economic and practical realities faced by local farmers, ensuring that our solution is both affordable and easy to use.

By combining hands-on field data with technical development, our goal is to create an innovative system that directly supports and uplifts rural mushroom farming communities in Sri Lanka. By combining technical development with real-world data from farmers, our group ensures that the final product is not only innovative but also applicable and beneficial to rural agribusiness in Sri Lanka.

8. Objectives and Novelty

Main Objective			
To design and develop a smart, IoT-based mushroom incubation system that enhances yield, ensures consistent quality, and promotes sustainable farming practices for small-scale oyster mushroom cultivators in Sri Lanka by enabling real-time environmental monitoring, automated control, and early detection of pests and contamination.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
M L Vidanaarachchi IT22063182	To design and implement a predictive mushroom fruiting system that combines real-time environmental sensing (CO_2, temperature, humidity, and time) with machine learning to forecast fruiting readiness.	▪ Collect real-time environmental data (CO_2, humidity, temperature, and time) from sensors installed in the cultivation environment. ▪ Log and preprocess collected data alongside actual mushroom fruiting times to create a training dataset. ▪ Develop and train a machine learning model that predicts the optimal time frame for mushroom fruiting. ▪ Integrate the prediction model into the control system to automate humidity and light adjustments ahead of fruiting.	This system uses machine learning to predict mushroom fruiting readiness from environmental data, enabling automated light control, a proactive improvement over timer- or rule-based methods.

		▪ Generate alerts via farmer interface to notify cultivators of upcoming fruiting phases for proactive preparation.	
G A I W Wijesinghe IT22062888	Smart Pest Detection and Automated Response System	▪ Design a mobile camera platform to patrol between mushroom rows, capturing images continuously from multiple angles. ▪ Apply real-time image processing and computer vision algorithms to detect pest presence or unusual movements on mushroom bags. ▪ Instantly alert farmers through a mobile/web app with pest location details. ▪ Integrate an automatic neem spraying mechanism that	This component uses a moving camera system to monitor pest activity across the mushroom farm, improving coverage and reducing blind spots. It automates pest control by linking real-time detection to instant alerts and automatic neem spraying, minimizing farmer intervention and contamination risk. The system also offers insights into pest patterns, helping identify hidden issues like unused zones or environmental problems early.

		activates only in affected areas to avoid overuse. ▪ Analyze pest frequency and location patterns to improve system accuracy and offer farm layout insights.	
WADUGE.S.W.O.P.K IT22586384	Predictive Advisory System for Moisture and pH Management in Mushroom Cultivation.	▪ Install moisture and pH sensors across different mushroom packet zones to monitor substrate health. ▪ Continuously collect and log sensor data to identify changing trends over time. ▪ Train a machine learning model to predict upcoming substrate imbalances or drying out periods. ▪ Develop an interactive dashboard that offers visual insights and personalized watering/pH advice. ▪ Incorporate farmer feedback into the model to retrain and improve prediction accuracy and relevance.	This system combines sensor data with machine learning to forecast moisture and pH trends and offer timely, AI-driven advice, transforming mushroom farming from reactive to proactive. Continuous learning via farmer input makes it intelligent and adaptive.

B H A S Peiris IT22290960	Investigating contamination causes and develop early detection and prevention strategies.	▪ Incorporate farmer feedback into the model to retrain and improve prediction accuracy and relevance. ▪ Train a CNN (Convolutional Neural Network) to detect contamination symptoms, such as mold, discoloration, or growth anomalies. ▪ Set up a real-time alert system that sends visual evidence (image + packet ID/location) to the farmer's app interface. ▪ Enable user feedback input via the app to confirm true positives and help refine detection accuracy. ▪ Retrain the detection model periodically with new data to improve sensitivity to early signs of contamination.	This system aims to detect and prevent mushroom contamination using a mobile camera system and CNN-based image analysis. It sends real-time alerts with images and packet locations via a farmer interface, allowing feedback and photo uploads. The system improves through adaptive learning, combining machine vision with IoT monitoring for accurate, automated detection.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
M L Vidanaarachchi IT22063182	This system includes environmental sensors to collect CO_2, temperature, and humidity data, with a data logger module to store readings for training a machine learning model that predicts fruiting readiness. A light control unit activates lighting at 75% intensity based on predictions or environmental triggers. The farmer interface provides alerts and visual insights, while an experimental setup compares prediction-based automation against traditional methods, aligning with IoT, machine learning, automation, UI/UX, and research evaluation specializations.
G A I W Wijesinghe IT22062888	This integrated IoT, and smart agriculture techniques to enhance pest control in mushroom cultivation. Using a moving camera image processing for real-time pest detection, combined with automated neem spraying and mobile alerts, the system reduces manual monitoring, prevents germ spread, and promotes efficient, data-driven farm management aligning directly with the objectives of intelligent farming technologies.
WADUGE.S.W.O.P. K IT22586384	This system uses sensors to monitor moisture and pH in real time, aligning with IoT and embedded systems. Machine learning models forecast future conditions, applying data science and time-series analysis. An advisory dashboard provides insights and suggestions, supporting UI/UX design. Farmer feedback is used to retrain the model, integrating adaptive learning and AI. Together, these components reflect specializations in IoT, machine learning, automation, and user-centered design.
B H A S Peiris IT22290960	The mobile camera system involves embedded hardware design; the CNN model applies machine learning for image-based disease detection; real-time alerts and IoT integration enable seamless communication and environmental monitoring; the farmer interface focuses on user interaction and usability; and the adaptive learning mechanism supports continuous AI model improvement through user feedback.

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Topic Assessment Form

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	✓	No	

b) Does the proposed topic exhibit novelty?

Yes	✓	No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes	✓	No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	✓	No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Proceed with minor changes.

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Topic Assessment Form

10. Supervisor's details

	Title	First Name	Last Name	Signature
Supervisor	Lecturer	Uditha	Dharmakeerthi	
Co-Supervisor	Assistant lecturer	Hanojhan	Rajahrajasingh	
External Supervisor	Lecturer	Pipuni	Wijesiri	
Summary of external supervisor's (if any) experience and expertise				

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Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.